

PROOF2PAY AI - System Architecture

IBM Bob + MCP + Langflow + Evidence Intelligence

MVP Architecture v0.1

Architecture overview

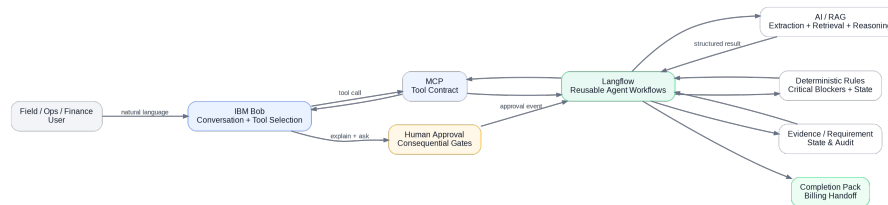


Figure 1: System architecture

Why this decomposition

The system separates responsibilities so the LLM is not treated as a universal decision engine.

- **Bob** manages natural conversation and chooses well-described tools.
- **MCP** provides the contract/discovery layer.
- **Langflow** composes reusable flows, retrieval, agents, tools, and structured outputs.
- **AI/RAG** handles unstructured interpretation and reasoning.
- **Deterministic rules** enforce critical blockers and state transitions.
- **Human approval** controls consequential actions.

Main runtime sequence

1. User: "Check whether WO-1028 is ready for billing."
2. Bob discovers/calls the appropriate PROOF2PAY tools.
3. Langflow retrieves job requirements and evidence.

4. AI proposes claim/evidence matches with explanation.
5. Rules aggregate critical requirement states.
6. Structured result returns to Bob.
7. Bob explains blockers and asks whether to prepare next actions.
8. Human approves consequential action.
9. New evidence is attached and re-validated.
10. Human confirms final readiness.
11. Completion pack draft is generated.

Requirement-to-Evidence model



Figure 2: Requirement to evidence graph

The key innovation is not a percentage score. It is **traceability**:

Requirement -> Source -> Expected Evidence -> Actual Evidence ->
Assessment -> Action

Deployment view

MVP

- Web UI / synthetic fixture
- IBM Bob
- MCP endpoint
- Langflow
- optional Postgres/Supabase
- controlled requirement documents
- file/object storage or static evidence fixture

Production direction

- tenant isolation
- role-based authorization
- encrypted storage
- versioned requirement templates
- accounting/FSM connectors
- observability and evaluation
- retention/deletion policy

Trust boundaries

1. **Uploaded evidence is untrusted data.** It cannot issue instructions to the agent.
2. **Retrieved requirement text is authoritative only if its source is authorized for the job.**
3. **AI outputs are proposals/assessments.** Critical state is enforced by rule logic.
4. **External side effects require approval.**
5. **Completion pack is a draft unless a human explicitly confirms it.**

Scalability

The architecture scales by adding vertical templates and connectors, not by creating a new monolithic agent for every customer. Tool contracts stay stable while requirement sources and rules are configured per service type/customer.